complete circuit of the test course. Sinusoidal variations are quite evident in the localizer tracking performance, but their amplitude generally decreased as range was reduced. Glide-slope control was considered satisfactorily precise, with the pilots tending to track high on the glide slope at the lower altitudes as an added safety margin. Airspeed control was generally poor and erratic, which indicated considerable control difficulty from the standpoint of both precision and accuracy.

A statistical analysis of all approaches, excluding the three aborts, was performed to determine the mean value and the standard deviation, or σ-value, associated with the localizer, glide-slope, and airspeed performance. The results of this analysis for a total of 69 approaches are presented in figures 11, 12, and 13 in terms of the mean value and the 3σ deviation. (Assuming a normal distribution for the deviations, the 3σ deviation would be exceeded only about 0.25 percent of the time.) Figures 11 and 12 show localizer and glide-slope performance, respectively, and include for comparison the deviation levels that would correspond to full-scale instrument deflection. The mean values and 3σ deviations computed at the Category I decision height for altitude, localizer, and airspeed error are summarized as follows:

	Mean	3σ
Altitude error	0.1 m (0.3 ft)	22.9 m (75.0 ft)
Localizer error	1.9 m (6.2 ft)	30.6 m (100.3 ft)
Airspeed error	4.1 knots	30.5 knots

Comparison of VFR and IFR Flight Times

A comparison of VFR and IFR flight times is presented in the following table for each leg of the route:

Route segments	Flight time, min		Time difference	
	VFR	IFR	Min	Percent
JFK to LGA	5.0	9.0	4.0	80
LGA to JRB	4.0	5.4	1.4	35
JRB to EWR	4.0	6.9	2.9	73
EWR to MMU	7.0	8.8	1.8	26
MMU to EWR	7.0	15.1	8.1	116
EWR to JFK	10.0	12.0	2.0	20